

Computer-Assisted Proofs for a Rational q, t -Catalan Formula

1 The theorem and the role of the computations

This note explains the computer-assisted part of the proof in Hawkes's paper *A conjectured formula for the rational q, t -Catalan polynomial* [1]. The paper proposes a symmetric-string formula for the rational q, t -Catalan polynomial. The formula remains conjectural in general, but the paper proves it through bounded deficit in the family

$$r \equiv 1 \pmod{s}.$$

Write

$$s = \ell + 1, \quad r = (\ell + 1)m + 1, \quad M_{\ell, m} = m \binom{\ell + 1}{2}.$$

The proved result is the following.

Theorem 1 (Hawkes). *If $r = (\ell + 1)m + 1$ with $m \geq 1$, then the conjectured symmetric-string formula is correct in every deficit layer*

$$0 \leq d \leq 20.$$

The computation is not a search for supporting examples. It verifies two finite obligations that the symbolic proof isolates. We call them *Computation A* and *Computation B*:

- Computation A verifies the proposed formula directly in the finite base range.
- Computation B verifies that the explicitly constructed canonical strings do not run past their prescribed endpoints.

The symbolic part of the proof then completes the canonical strings at one critical length and transports the resulting decomposition to every larger length.

More generally, the paper shows that for any fixed cutoff d^* , producing this kind of combinatorial proof for all layers $d \leq d^*$ in the family $r \equiv 1 \pmod{s}$ is equivalent to a finite verification problem. The choice $d^* = 20$ is therefore not intrinsic to the method. A larger cutoff would require additional finite calculations, which have not been recorded here.

2 The proposed strings

Let $D_{\ell, m}^d$ denote the position-coordinate m -Dyck paths in deficit layer d , and let

$$C_{\ell, m}^d(q, t) = \sum_{x \in D_{\ell, m}^d} q^{\text{area}(x)} t^{M_{\ell, m} - d - \text{area}(x)}.$$

The distinguished roots form a subset $T_{\ell, m}^d$. In the position coordinates used by the code, a path is a sequence

$$x = (a_0, a_1, \dots, a_\ell), \quad a_0 = 0, \quad 0 \leq a_{i+1} \leq a_i + m,$$

and the roots in $T_{\ell, m}^d$ are characterized by beginning

$$(0, 0, \dots).$$

For integers a, b , define the signed symmetric interval

$$\text{sym}(a, b) = \frac{q^{b+1}t^a - q^at^{b+1}}{q - t}.$$

When $a \leq b$, this is the ordinary positive string

$$q^at^b + q^{a+1}t^{b-1} + \dots + q^bt^a.$$

The conjectured layer formula is

$$C_{\ell, m}^d(q, t) = \sum_{x \in T_{\ell, m}^d} \text{sym}(\text{area}(x), M_{\ell, m} - d - \text{area}(x)). \quad (1)$$

Thus every root prescribes a “string holster”: one slot at every consecutive area from the root area to its symmetric terminal area.

When $d < (\ell - 1)m$, the paper constructs the roots from fitting partitions. If $P_{\ell-1}^d$ is the set of partitions of d whose largest part is at most $\ell - 1$, then maps g and f give a bijection

$$f \circ g : P_{\ell-1}^d \longrightarrow T_{\ell, m}^d.$$

For $m = 1$, once $\ell - 1 \geq d$, these fitting partitions are simply all partitions of d .

3 Canonical initial strings

For each fitting partition λ , let

$$x_\lambda = f(g(\lambda)).$$

The paper defines a local operation right. Beginning at x_λ , repeatedly apply this operation until it is no longer possible:

$$x_\lambda, \text{right}(x_\lambda), \text{right}^2(x_\lambda), \dots, \text{lowest}(x_\lambda).$$

Each step raises area by one while preserving the deficit. These are the *canonical initial strings*.

The following facts are proved symbolically in the paper:

- distinct canonical strings are disjoint;
- the maps left and right are inverse where they are defined;
- the canonical strings partition the subset $C_{\ell, m}^d \subseteq D_{\ell, m}^d$ of connected paths.

The remaining paths

$$N_{\ell, m}^d = D_{\ell, m}^d \setminus C_{\ell, m}^d$$

are available to complete the canonical strings. The added tails need only occupy consecutive area levels; they are not required to continue the right operation.

4 The critical length

Fix a cutoff d^* . For each m , define

$$\ell^*(m, d^*) = \min\{\ell : (\ell - 1)m > d^*\}.$$

The proof has three stages:

1. verify the formula directly for $\ell \leq \ell^*$;
2. prove that the canonical strings at $\ell = \ell^*$ can be completed;
3. propagate that completed decomposition symbolically to every $\ell > \ell^*$.

For the theorem, $d^* = 20$.

5 Computation A: the finite formula check

Computation A is Lemma 3 of Section 9; it is called “computation 2” in the paper’s appendix. For each required pair (m, ℓ) , it verifies equation (1) for all generated deficit layers $d \leq 20$.

The code represents the monomial

$$q^j t^{M_{\ell,m} - d - j}$$

by the pair $(j, M_{\ell,m} - d)$. It creates three multisets:

- *all*: one record for every generated path;
- *plus*: all positive string slots prescribed by the roots $a_1 = 0$;
- *minus*: the correction slots coming from roots whose signed interval is oriented past its midpoint.

It then checks the exact multiset identity

$$\text{plus} = \text{all} + \text{minus}. \quad (2)$$

This is a direct coefficient-by-coefficient verification of the conjectured formula in the finite base range

$$0 \leq d \leq 20, \quad \ell \leq \ell^*(m, 20).$$

Computation A has two logical uses. First, it proves the theorem outright for all $\ell \leq \ell^*$. Second, at $\ell = \ell^*$, it says that the actual paths occur at every area with exactly the multiplicity required by all the proposed full string holsters.

In `code.py`, this is implemented by `check_lemma3`; the exposition-friendly alias is `check_computation_a`.

6 Computation B: no canonical string overshoots

Computation B is Lemma 2 of Section 9; it is called “computation 1” in the paper’s appendix. It is needed only at the critical length ℓ^* .

Let $\lambda \in P_{\ell^*-1}^d$, and let $h(\lambda)$ be the number of parts of λ . The complete string indexed by λ is supposed to end at area

$$R(\lambda) = M_{\ell^*,m} - d - h(\lambda).$$

Computation B constructs the canonical endpoint

$$\text{lowest}(f(g(\lambda)))$$

and verifies

$$\text{area}(\text{lowest}(f(g(\lambda)))) \leq R(\lambda). \quad (3)$$

Thus the canonical string may be too short, but it is never too long. If it passed its prescribed endpoint, the proof would have to shorten and regroup a rigidly constructed string. Inequality (3) rules out that obstruction.

The program recognizes the roots by the condition $a_1 = 0$, computes their terminal paths with `lowest`, and recovers $h(\lambda)$ through the source statistic named `height`. In `code.py`, the check is `check_lemma2`; the exposition-friendly alias is `check_computation_b`.

7 Why the two computations complete the boundary strings

At $\ell = \ell^*$, each canonical string fills a disjoint initial segment of its prescribed holster. Computation B proves that no initial segment extends beyond the end of its holster. Computation A proves that, at each area level, the total number of actual paths equals the total number of holster slots.

Subtract the slots already occupied by the canonical strings. At every area, the number of unused paths is then exactly the number of unfilled slots. Since an extension tail is not required to follow right, the unused

paths can be assigned area by area to the open slots. The result is a partition of all paths into completed strings with the prescribed starts and ends. In the terminology of the paper, the layer is *extendable*.

The construction is rigid in its canonical prefixes but flexible in its added tails. Correct multiplicities alone would prove the formula at the critical length, but the canonical prefixes are essential for the symbolic propagation to larger ℓ .

8 Symbolic propagation

Once every layer $0 \leq d \leq d^*$ is extendable at ℓ^* , Claim 16 of the paper proves symbolically that every such layer is extendable for all

$$\ell > \ell^*.$$

The induction removes the largest parts from the partition indexing a root, uses the extension already constructed one length lower, and prefixes the appropriate new coordinate. The proof checks symbolically that areas, deficits, endpoints, and the unused-path partition all transform correctly. No further computation is used after the critical length.

Consequently,

Computation A below and at ℓ^* + Computation B at ℓ^* + symbolic propagation

proves the conjectured formula for every ℓ .

9 Example: $m = 3$

Take $m = 3$ and $d^* = 20$. The critical length is

$$\ell^* = 8,$$

because $7 \cdot 3 > 20$, while $6 \cdot 3 \leq 20$.

The proof specializes as follows.

1. **Finite base range.** Computation A verifies the formula for every $0 \leq d \leq 20$ and every $\ell \leq 8$.
2. **Canonical strings at the boundary.** For each fitting partition of each d , form $f(g(\lambda))$ and apply right until the path is unrightable. The symbolic theory proves that these canonical strings are disjoint.
3. **No overshooting.** Computation B checks that every canonical endpoint lies at or before its prescribed terminal area.
4. **Complete the tails.** Computation A supplies the exact area multiplicities needed to allocate all leftover paths to the open slots. Hence every layer at $\ell = 8$ is extendable.
5. **Propagate.** Claim 16 proves the formula for every $\ell > 8$ without further computation.

10 The finite run and its scope

For $d^* = 20$, the appendix computation checks $1 \leq m \leq 20$. For $m > 20$, the critical length is $\ell^* = 2$, and the paper handles the remaining cases directly by symbolic arguments.

The recorded full run generated 5,692,942 source-relevant path records. Computation B found no failures. Computation A checked 106 nontrivial (m, ℓ) -layers and found no failures. The stable summary is

```
lemma2_status: PASS, lemma2_failures: 0,
lemma3_status: PASS, lemma3_layers_checked: 106, lemma3_failures: 0.
```

The readable file `code.py` closely follows the appendix routines and uses only the Python standard library. Its source defaults are `--max-m 20` and `--dstar 20`. Smaller checks can be run, for example, with

```
python code.py --max-m 3 --dstar 6.
```

The options `--check A` and `--check B` select the two computations in the terminology of this note; `--lemma 3` and `--lemma 2` are equivalent source-paper labels.

The theorem established here concerns the family $r \equiv 1 \pmod{s}$ through deficit 20. The unrestricted formula for arbitrary coprime (r, s) , and the same family in deficits above 20, remain conjectural.

References

- [1] Graham Hawkes, *A conjectured formula for the rational q, t -Catalan polynomial*, Annals of Combinatorics **28** (2024), 749–795; arXiv:2208.00577.